

IBM BOB 2.0 HACKATHON

Orient

A smart developer-onboarding assistant that turns days of codebase archaeology into minutes.

Team **minahiru** · Built with IBM Bob 2.0 · lablab.ai

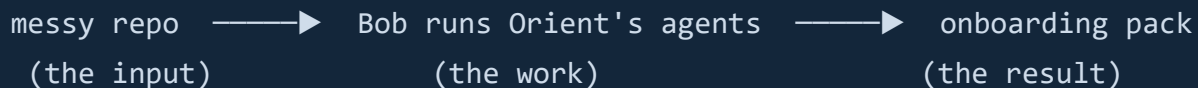
A new developer loses **days** before their first useful change.

Not too hard problems — to archaeology that repeats for every hire:

- The README's setup steps don't work; required config is undocumented
- The docs describe the system as designed, not as it runs today
- It's unclear which code is live and which is long-dead
- Nobody has assembled what a newcomer actually needs to know

THE IDEA

Point **Orient** at an unfamiliar codebase and it produces a **verified onboarding pack** — the real setup steps, a map of the code, and a list of everywhere the docs lie.



A small pipeline of focused agents

01

Env Archaeologist

Finds every environment variable the code actually reads. Outputs a complete **.env.example** and a real **SETUP.md**.

02

Doc Reconciler

Compares the README to the code. Outputs **DOC-CORRECTIONS.md** — every contradiction, with file:line proof.

03

Verifier

A fresh agent uses **only the pack** to set the app up and pass a failing test. Proof, not a claim.

Each agent runs as a focused Bob subagent, so one investigation never clutters another.

Bob does the multi-step work — not just the coding

BOB CAPABILITY	HOW ORIENT USES IT
Agent mode	Reads the repo and writes the onboarding-pack files directly
Subagents	Each investigation runs in its own isolated context
Skills	Env-archaeologist & doc-reconciler are reusable, repeatable workflows
Document understanding	Reconciles stale prose docs against real source code

Same repo. Two experiences.

FOLLOWING THE README

```
$ node app.js
Error: Cannot find module 'app.js'

$ npm start # after guessing
Error: Missing required environment
variable: JWT_SECRET
    at required (src/config.js:8)

# app never starts
```

WITH ORIENT'S PACK

```
$ cat onboarding-pack/SETUP.md
# real steps + complete .env.example
$ npm start
TaskFlow API listening on port 8080

# running on the first try
# DOC-CORRECTIONS.md lists 6 doc errors
```

THE PROOF

A fresh Bob agent, with no memory of the earlier work, used **only the generated pack** to run the app and make a failing test pass.

That turns "the output looks good" into "the output demonstrably works."

From "never starts" to "runs on the first try."

OUTCOME	FOLLOWING THE README	WITH ORIENT
Does the app start?	No — 2 blocking crashes	Yes, first attempt
Wrong / missing doc claims	6, silently misleading	All 6 caught, with file:line
Undocumented env vars	3 (app dies on one)	All 3 found + documented
Failing test	Still failing	Made to pass

Measured on the sample service in this repo.

Where it stops — and where it goes

LIMITS TODAY

- Tuned for a single-language service; a monorepo runs per-package
- The pack is a snapshot — as current as the code at run time

NEXT

- Read live docs (wikis, Confluence) via MCP, not just local files
- Run on every PR to keep the pack fresh automatically

THANK YOU

Orient

Turning codebase archaeology into a five-minute head start.

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github.com/minahiru2007/orient-hackathon